COURSE TITLE : QUALITY ASSURANCE AND TESTING
COURSE CODE : 6092
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/6
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Standards and standard organisations for raw material testing	20
2	Testing of ISNR and concentrated latex	20
3	Physical testing of rubber	20
4	Processibility tests and Product testing To understand the statically control charts and analysis of test data	15
TOTAL		**75**

COURSE OUTCOME

Sl.	Sub	Student will be able to
1	1	To understand the various standard organisations and their significance in quality control of polymer products. To understand the various raw material testing
	2	To know the various test methods employed to study the thermal, electrical, and optical properties of polymers.
2	1	To understand the specifications and test methods for ISNR To understand the specifications and test methods for concentrated latex.
	2	To understand the various testing equipments and test methods employed for plastic and rubber
3	1	To understand the various process ability tests used in rubber & Plastic processing
	2	T o understand the various specification tests for rubber products To understand the statically control charts and analysis of test data.

SPECIFIC OUTCOME

MODULE I

To understand the various standard organizations and their significance in quality control of polymer products.

1.1.0 To understand the various raw material testing

1.1.1Define quality

1.1.2Discuss the quality concepts in modern life

1.1.3Know the various world standard organization and their role in standardization and quality assurance – ISO, BIS, ASTM, DIN

1.1.4Know the procedure for accreditation of ISI certification

1.2.0 Raw material testing

1.2.1 Testing of important additives-% purity of DPG,MBT&ZnO. -Moisture content of clay and Silica.

MODULE II

To know the various test methods employed to study the thermal, electrical, and optical properties of polymers

2.1.0 Thermal properties

2.1.1Thermal conductivity

2.1.2 Flammability

2.1.3 V.S.T

2.1.4 Melting point

2.1.5 Specific heat

2.1.6 Brittleness temp

2.2.0 Electrical properties

2.2.1 Dielectric strength

2.2.2 Dielectric constant

2.2.3 Arc resistance

MODULE III

To understand the specifications and test methods for ISNR.
To understand the specifications and test methods for concentrated latex.
To understand the various testing equipments and test methods employed for plastic and rubber

3.1.0 Describe the principle, procedure and significance of various tests of dry rubber (ISNR) & concentrated latex

3.1.1Describe the method for the determination of dirt content, volatile matter, ash, Nitrogen content, Po ,PRI ,Ash , and colour

3.1.2Compile the technical specifications of ISNR as per BIS

3.1.3State the significance of each specification parameter on the quality improvement of ISNR

3.1.4Describe the significance, principle and method for the determination of total solid content, dry rubber content, alkalinity as ammonia, VFA Number, KOH Number, MST, Mg content, viscosity, density, coagulum content, sludge content & Mn,

3.1.5Compile the technical specification of concentrated latex as per BIS

To understand the various process ability tests used in rubber & Plastic processing
T o understand the various specification tests for rubber products
Describe the significance, principle, equipments used and methods of testing of plastics and rubbers.

3.2.0Explain sample preparation and principle of conditioning

3.2.1Discuss the various tests relevant to plastics and rubbers

3.2.2Define TS, M_{300} and EB

3.2.3Describe the test method for determining TS, M_{300} and EB

3.2.4Define tear strength and explain the method for its determination

3.2.5Explain permanent set

3.2.6Define compression set and tension set and describe the methods of determination

3.2.7Explain the significance of compression set in Engineering applications of rubber

3.2.8Define hardness, abrasion resistance, resilience and explain the procedure for its determination

3.2.9 Study briefly the significance, principle and method of determination of flex cracking fatigue failure

3.3.0 Processability Test

3.3.1List out the various testing equipments used for process ability tests

3.3.2Explain the cure-characteristics of compounded rubber with the help of cure curves obtained using Mooney Viscometer, ODR and MDR

3.3.3 Determination of melt flow index

<u>MODULE IV</u>

4.1.0 Understand the specification and test methods for dry rubber and latex products

4.1.1State the importance of product testing

4.1.2Give the IS specification of important rubber products

4.1.3Describe the various test methods & specifications for cycle tyretubes, M.C. soles, Hawaii, V-straps, LPG tubing, & conveyor beltings
4.1.4Describe the specifications and test methods for latex foam
4.1.5Describe the specifications and test methods for surgical, & examination gloves.

4.2.0 Testing of Plastic product

4.2.1 PVC pipe-Blood bag
4.2.2 Water tank,
4.2.3 Disposable syringes-
4.2.4 Helmet- Impact test- Charpy and Izod

4.3.0 To understand the statistical control charts and analysis of test data

CONTENT DETAILS

MODULE – I

Introduction and material testing

Importance of quality assurance and Testing-Quality – definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organization. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction.
Raw material testing-Testing of Additives-% Purity of DPG, MBT, ZnO, moisture content of clay, silicaThermal properties Thermal conductivity, flammability, vicat softening temperature, M.P, brittleness, sp. heat, HDT-Electrical properties **-**Dielectric strength, dielectric constant, arc resistance

MODULE-II

Specification test for ISNR and latex concentrate

Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N_2 content, Po and PRI,ASHT and Colour-Technical specifications of ISNR as per BIS-Specification test for concentrated latex-
Significance, principle and methods of the following tests – KOH No, VFA number mechanical stability time, Mg content, sludge and coagulum content, sp gravity, viscosity, TSC, DRC, Alkalinity etc. Technical specifications for conc. Latex as per BIS

MODULE – III

Physical properties

Method of testing, specimen standards-principle of conditioning-equipment used-significance of following tests relevant to plastics and rubbers – TS, tear strength, tension set and compression set, hardness, Abrasion resistance, resilience, flex cracking, impact strength, fatigue failure.

Importance of process ability tests-Principle, method of testing, equipments used for cure – characteristic study-Illustration with graphs and advantages of cure meters-Different types of cure meters – Mooney viscometer, Rheometer, recent developments in Rheometer- Determination of melt flow index, bulk factor and shrinkage of plastic materials

MODULE-IV

Polymer product testing

Testing of dry rubber products-Importance of product testing- IS specifications and specific test procedures of cycle tyre tubes, M.C. soles, Hawai V-straps, LPG tubings, conveyor beltings-

Testing of latex products-ASTM/IS specifications and specific test procedures of latex foam, surgical gloves, examination glove. Testing of Plastic product - PVC pipe- blood bag-water tank, disposable syringes- helmet- Impact test-Charpy and Izod

Stastical control charts U, P, mean, deviation ,correlation &interpretation of data.

REFERENCE BOOKS

1. Rubber Technology and manufacture – C.M. Blow
2. Physical testing of Rubbers – Scott
3. Physical testing of Rubbers – R.P. Brown
4. Indian standards
5. ASTM standards – Vol. 8 & 9 series
6. Plastic testing technology – Vishu Sha
7. FRP Technology – Weather head